

Harnessing the potential of zinc oxide nanoparticles and their derivatives as nanofertilizers: Trends and perspectives[☆]

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ABSTRACT

The agriculture sector is currently facing a decline in plant productivity and yield. Different technologies are being developed to combat these risks. However, innovation in existing technologies is required. Nanotechnology has the potential to solve these difficulties by modifying traditional farming practices and approaches. Nanoparticles (NPs) can bind and transport various substances, such as plant nutrients, and govern their slow release over a longer period, which can reduce the danger of nutrient losses while maintaining environmental safety. In this aspect, the role of zinc oxide nanoparticles (ZnO NPs) and their derivatives in agriculture has recently sparked a lot of interest. ZnO NPs can be coated with different compounds which enhance their biocompatibility within the plant cells. The unique nanostructures and nano-characteristics of ZnO NPs and their derivatives have resulted in the development of a novel approach for boosting plant development and productivity as well as improved stress tolerance via targeted delivery and slow-release mechanism, resulting in enhanced nutrient use efficiency, regulating phytohormone levels, enhancing root morphology, and increasing enzyme activity, leading to their application as nanofertilizers. There are important knowledge gaps regarding the long-term environmental consequences and the specific biochemical pathways influenced by ZnO NPs. This review aims to provide an overview of the most recent advancements in the use of ZnO NPs in agriculture, identify areas where more research is needed, and suggest potential future research directions.

Introduction

Agriculture is a crucial component of the economy that produces a variety of crops for various food and feed demands (Yaqoob et al., 2022). In the current time, the agriculture sector is facing challenges due to various factors like climate change. These challenges are reducing productivity and yield of plants (Skendzić et al., 2021). For decades, extreme changes in the environment including increased temperature, drought, salinity, floods, etc., are hindering the development in agriculture (Pawlak and Kołodziejczak, 2020). Different forms of biotic and abiotic stresses are reducing the overall plant production (Thounaojam et al., 2021). Apart from this, the global population is increasing with time. In the 1950s, it was 2.5 billion and the number increased up to 7.2 billion in 2014. According to an estimation, the global population will increase up to 9.6 billion by 2025. This high population increase demands double the amount of food being produced by agricultural means.

It puts pressure on growers to boost agricultural production (Sekhon, 2014). Another challenge is the high rate of urbanization, which is reducing the area of agricultural land. As a result, the fertile land area is running out and the amount of food production is not enough for rapidly growing population (Naderi and Danesh-Shahraki, 2013). Moreover, the mortality rate caused by food scarcity is expected to be higher than that of illnesses such as Acquired Immunity Deficiency Syndrome (AIDS), Tuberculosis (TB), and malaria (Rohr et al., 2019). To meet food demand for a huge population within limited resources, it is required to enhance the food production from the same area of available land (Wang, 2022).

Multiple technologies are being developed to overcome these threats. But innovation in the existing technologies is necessary that includes improvement in conventional fertilizers, production of new pesticides, etc. (Chávez-Dulanto et al., 2021). Among the emerging technologies, nanotechnology can provide solutions to these problems by the modifications of conventional practices and techniques. It can

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significantly improve the yield and production of plants. Thus, the field of nanotechnology has a dominating position in the transformation of agriculture (Ashraf et al., 2021). One of the practices to improve the production from plants is the application of novel agrochemicals (Javed et al., 2023a). Due to various factors like drifting, leaching, photolysis, hydrolysis, and microbial degradation, the applied chemicals cannot reach the target site. For the efficient and site-specific delivery of agrochemicals, the use of nanoparticles (NPs) and nano capsules is an advantage. Field application of NPs is also gaining attention in recent times for the slow and effective delivery of fertilizers, pesticides, and herbicides (Avila-Quezada et al., 2022; Prasad et al., 2012).

Recently, there has been a lot of interest in the role of ZnO NPs in plant biotechnology. NPs of ZnO gained attention for their use in agriculture due to their unique properties. ZnO NPs with fundamental sizes ranging from 1 to 100 nm are some of the frequently employed NPs in nanotechnology. The distinctive nanostructures and nano-characteristics of ZnO NPs have enabled them for increasing plant development, production, and yield (Thounaojam et al., 2021). ZnO NPs offer advantages like, (a) increase of seed germination and seedlings development by an increased absorption of Zn ions (Hanif et al., 2023), (b) mitigation of abiotic stresses by an increased enzymatic and non-enzymatic antioxidant levels along with the boost of photosynthesis (Hanif and Zia, 2023; Szerement et al., 2022; Azam et al., 2022), (c) improvement of nutritional quality of plants by modifying carbohydrate, protein, and mineral content, and (d) limitation of the growth of pathogenic microorganisms for plant's resilience and minimizing the pesticide input.

Previously, ZnO NPs led to a better impact on the photosynthetic efficiency of crops as compared to bulk ZnO (Azarin et al., 2023). ZnO NPs have been used as nanofertilizers to alleviate various kinds of stresses faced by plants, including heavy metals, salinity, drought, high and cold temperatures, and mainly nutrient stress while positively effecting the plant yield (Khan et al., 2023; Kumar et al., 2021). Seed priming with ZnO NPs has been reported to overcome cobalt (Co) toxicity in maize. Reactive oxygen species (ROS) and MDA accumulation was found to be reduced with a decreased Co uptake, with a higher nutrient accumulation, and rendered stability to maize plant photosynthetic apparatus and ultracellular structures (Salam et al., 2022). ZnO NPs alleviated drought stress by increasing plant water uptake by upregulating the transcription of aquaporin genes, thus promoting seed germination (Liu et al., 2022). ZnO NPs have been reported to increase photosynthetic pigments, seed germination, and the yield of wheat by upregulating the activities of nutrient mobilizing enzymes (Liu et al., 2022; Rai-Kalal and Jajoo, 2021). ZnO NPs mitigated negative impacts of salinity stress on tomato and boosted its protein content and antioxidative activity like superoxide dismutase (SOD), catalase (CAT), and peroxidase (POD) (Faizan et al., 2021).

To the best of our knowledge, there is no review article previously published in which the role of ZnO NPs as nanofertilizers has been comprehensively exploited. Moreover, the derivatives of ZnO NPs behaving as novel nanofertilizers have been discussed for the very first time in a review study. The goal of this review article is to compile the recent studies performed on the *in vitro* and *in vivo* applications of ZnO NPs and their derivatives on different plants. It analyzes their beneficial effect as nanofertilizers for boosting the yield, growth, and phytonutrients of respective plant species, with an insight on overcoming abiotic stresses faced by plants. It offers comprehensive information that can guide future innovations in this field. Hence, addressing the potential of ZnO NPs and providing insights into optimization of nanofertilizers, this article stimulates further research in exploiting ZnO NPs and their derivatives for crop sustainability.

Role of nanotechnology in agriculture

Agriculture has been one of the major contributors to the ecological and economic success of countries and more than half of the human

population depends upon it for their livelihood. There are multiple constraints faced by agricultural sector such as decline in soil's organic matter, low water availability, nutrient deficiency, and climate change. Sustainable agriculture is the only way to overcome food insecurity and achieve "zero hunger" which is one of the major sustainable development goals stated by United Nations (Arora and Mishra, 2022). This challenge could be dealt with advanced technologies along with modifications in the global food production system.

Thus, studies on role of nanotechnology and its promising potential in agricultural sector has gained a lot of attention during the last years. It can revolutionize the agro-food industry by innovative tools such as nanofertilizers, nanopesticides, nanobiosensors, feed additives, and remediation of contaminated soil via nanomaterials. The role of NPs in the sustainability of agriculture has been given in detail in Fig. 1.

Presently, nano-techniques are focused to increase efficiency and stability, planned delivery, and controlled release of agrochemicals (Singh et al., 2022). Nanosensors and nanocides also help in fighting plant viruses and other crop pests. It has been reported that usage of nanocides in the agriculture sector has led to a decline in dosage of conventional pesticides, therefore, saving humans from exposure to these harmful chemicals. Nanoremediation eradicates contaminants from soil and irrigation water due to the highly specific features of NPs such as high surface area and effective reactivity. However, the response to NPs varies according to the plant's species, environmental factors, and NPs exposure duration. So, it is imperative to carry out screening and optimization of nanomaterials before making them commercially available (Usman et al., 2020).

NPs as nanofertilizers

In the case of conventional/chemical fertilizers, the supply of nutrients in chemical form cannot be readily absorbed by plants. Furthermore, most macronutrients applied in this way are poorly soluble in soil. Consequently, regular supply of these nutrients becomes necessary (Kumar et al., 2020). Hence, there is need to develop a type of fertilizer that can overcome the challenges posed by conventional fertilizers. As mentioned earlier, there is a surge in population and food demand, hence, to meet this demand, agricultural productivity must be increased. There is need for low-cost long-term technology that can supply nutrients resulting in increased crop production with lower resource consumption.

A new class of fertilizer is nano-based fertilizer. Nanofertilizers offer controlled and slow release of nutrients as compared to that of conventional fertilizers, making them efficient and cost-effective (Yadav et al., 2023). Due to the high surface to volume ratio property of nanofertilizers, they are suitable in withholding and slow release of nutrients (Gade et al., 2023). These fertilizers can reduce the leaching and runoff of nutrients into the environment (Khatri and Bhateria, 2023). For an adult tree, 150 kg of ordinary fertilizer is required, whereas the amount can be reduced to 40–50 kg using nanofertilizer (Bedi and Kaur, 2015; Raliya et al., 2016).

Moreover, the use of nanofertilizers in terms of increase in productivity of crops while reducing the stresses has shown encouraging results (Waleed, 2023). Indeed, nanotechnology has significant advantages over conventional fertilizers in terms of productivity and enhanced nutrient availability. Hence, greater application rate efficiency, as well as focused administration of certain micronutrients, slow release, fewer leaching losses, and increased yield capacity makes NPs suitable to be used as nanofertilizers (Gade et al., 2023) (Fig. 2).

Based on chemical nature, NPs can be classified into two types: organic NPs (carbon-based) and inorganic NPs (metal-based). Metal and metal oxide NPs are important for the growth of plants as nutrients with several advantages (An et al., 2022). Organic NPs include polymers, lipids, carbon nanotubes (CNTs), graphene, etc. (Anandhi et al., 2020; Selyutina et al., 2017). Based on the type of nutrient and carried compound, there are various types of nanofertilizers such as micronutrient-

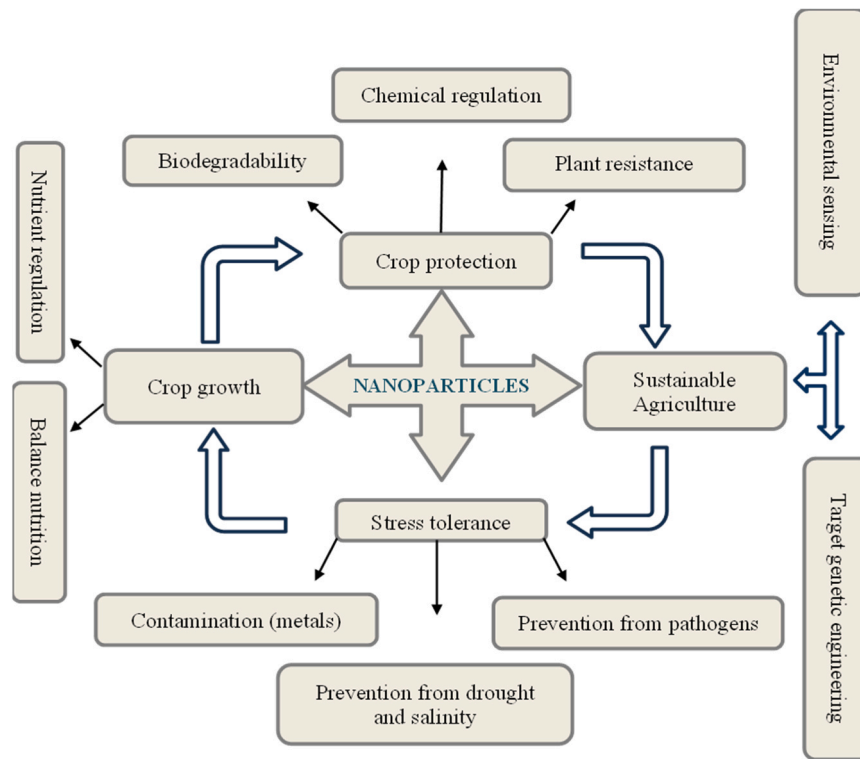


Fig. 1. Use of nanotechnology and nanoparticles in agriculture.

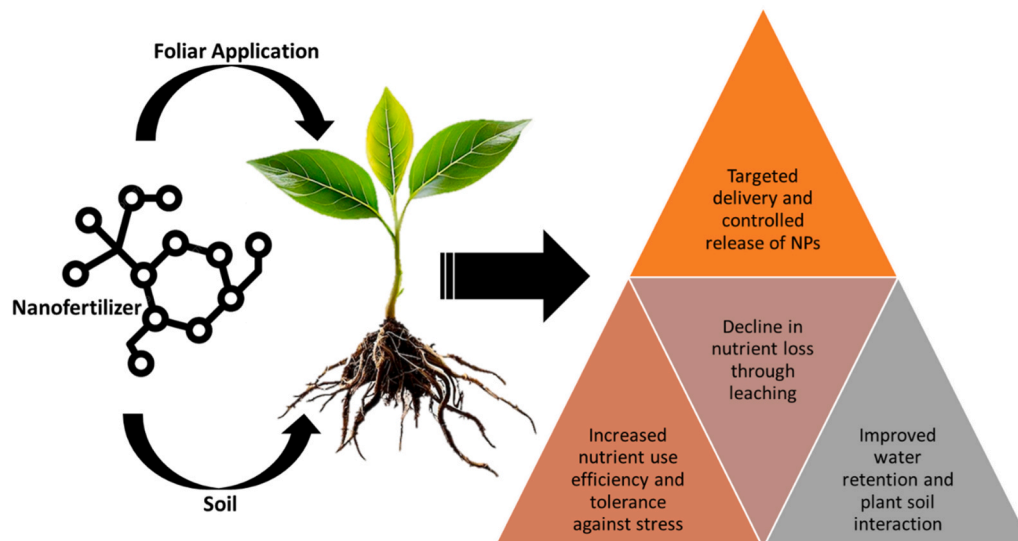


Fig. 2. Benefits associated with soil and foliar supplementation of nanofertilizers.

macronutrient-, polymer-, zeolite-, and trace element-based nanofertilizers (Gade et al., 2023). Micronutrients are required in small amounts for different processes in plants. These are Cu, Fe, Zn, Mn, Co, Cl, Mo, etc. (Chrysargyris et al., 2022). Fe is required in electron transfer system, chlorophyll synthesis, and enzymes functions. Zn is required for different metabolic enzymes for catalytic purposes, cell division, protein synthesis, and maintenance of structure and function of plasma membrane. Zn insufficiency is common around the world due to its poor availability in vegetable-based diets. Mo is required for nitrate reductase enzyme and acts as an important element in nitrogen fixation in legume crops. Cu is involved in various physiological activities like cellular transport, anti-oxidation, hormone signaling, and mitochondrial

respiration, etc. Co maintains cellular homeostasis and Cl functions in osmotic and stomatal regulation (Zewide and Sherefu, 2021).

Among the major macronutrients (N, P, K), Nitrogen (N) has a vital role in protein synthesis and energy metabolism in plants. Plants can only absorb N in the form of nitrate and ammonium ions (NO_3^- & NH_4^+). Negatively charged nitrate (NO_3^-) is not readily absorbed by the soil due to its poor affinity for soil particles, hence it is lacking there. N combined with hydroxyapatite and zeolite enhances N uptake in soil while slowing its release (Preetha and Balakrishnan, 2017; Bernela et al., 2021). Potassium (P) is vital for storage and transportation of energy, photosynthesis, root growth, flowering, ionic balance, and many other processes in plants. P is known to trigger the activity of up to sixty plant

growth promoting enzymes (Bernela et al., 2021). Potassium (P) is essential to fulfill critical functions of osmotic balance, enzymes activation, pH homeostasis, and membrane electric potential regulation (Sardans and Peñuelas, 2021). Calcium (Ca)-based nanofertilizers have a role in cell wall stability, mineral transport, retention of minerals in soil, detoxification of toxic substances, and seed formation, etc. Magnesium (Mg) is present in the core of chlorophyll molecules and activates enzymes. Sulphur (S) is involved in the formation of chlorophyll; it also enhances N availability and is involved in plant defense mechanisms. Moreover, S also produces few amino acids (Sachan et al., 2021 and Kumar et al., 2022).

A very small amount of trace elements (minerals or metals) is present in the living tissues. Some of these elements have importance in agriculture (Heinisch et al., 2019; Liu et al., 2017). Nanofertilizers based on trace elements have demonstrated encouraging results in terms of increasing crop growth and production while decreasing nutrient loss and environmental threats (Deepika et al., 2020). Nanoscale materials employed as crop nutrients have been shown in studies to improve germination, growth, nutrient absorption, and nutrient bio-fortification in different crops (Yuvaraj and Subramanian, 2020). Furthermore, low concentration of iron-based NPs has been shown to improve root development in legumes allowing for a more ecologically friendly method to fertilization (Singh et al., 2021). These findings suggest that nanofertilizers based on trace elements might be a reliable strategy for increasing agricultural yield while minimizing the environmental impacts.

Zeolites are naturally occurring minerals containing nutrients for plants. These agents gradually release nutrients. Nanofertilizers based on zeolites improve agricultural nutrients availability to crop plants across the growth cycle. The gradual release of nutrients can prevent loss of nutrients in the soil (Jakhar et al., 2022). Nanozeolite and its combined forms can control N release due to their greater surface area. Thus, they have been frequently used in the development of nanofertilizers (Mahanta et al., 2019). In the light of this, many recent studies have explored the use of nanozeolites as nanofertilizers and nanocarriers (Sharma et al., 2022). Application of nanozeolites improved the different growth parameters of maize (Chaudhary et al., 2021), wheat (Kheir et al., 2018), lettuce (Khan et al., 2021), and many more crop plants.

Different polymers like alginate, chitosan, albumin, etc., have also been studied for the delivery of NPs to plants. Among these, biopolymer chitosan is inexpensive and can function in the growth of plants, nutrient uptake, germination of seeds, photosynthesis, and crop yield (Malerba and Cerana, 2019). Furthermore, antimicrobial properties of chitosan are reported (Kumaraswamy et al., 2018). Studies also documented the combined effect of polymers and NPs. For example, application of Zn-chitosan NPs improved the growth of wheat crop (Deshpande et al., 2017). Similarly, Cu-chitosan NPs were used in increasing the growth of tomato and maize plants (Saharan et al., 2015; Saharan et al., 2016).

Role of Zn in plants

Zinc (Zn) is a plant micronutrient and essential trace element that carries out various physiological and biochemical functions in plants. It is a key component of enzymes that catalyze multiple plant metabolic reactions and plays a structural and catalytic role in over 300 proteins. Zn fortifies plants' defense mechanisms, lessens disease severity, enhances resistance to drought stress, reduces heavy metal toxicity, and promotes overall growth and yield (Broadley et al., 2007). It is vital for plant resistance against diseases, necessary for protein synthesis, pollen formation, cell membrane integrity, and enhances chlorophyll and antioxidant enzyme levels in plant tissues. Zn operates in numerous enzymes and proteins that are crucial for plant response to pathogens, such as alcohol dehydrogenase (ADH) and superoxide dismutase (SOD), which are essential for maintaining cellular homeostasis and safeguarding plants from oxidative damage caused by (ROS) (Cabot et al.,

2019). The nutritional status of the host plant affects pests and pathogens' colonization success, and micronutrients like Zn, Cu, Mn, and Fe play a significant role. Zn is a major immune response regulator in both plants and animals. Both Zn deficiency and excess cause Zn to act as a prooxidant, so its concentration and compartmentation are tightly regulated by different proteins (Cabot et al., 2019). Agricultural soils typically contain 10–300 µg of Zn per gram of soil. However, approximately half of the soil is naturally low in Zn due to various soil deformities and human activities (Hacisalihoglu, 2020). For most crops, the normal Zn level ranges from 30 to 200 µg Zn per gram of dry weight (DW). The application of Zn in plant culture has been demonstrated to exhibit a range of benefits, including the mitigation of metabolic disorders and toxic effects that may arise from excessive concentrations of Zn per gram of dry weight. On the other hand, when applied in appropriate concentrations, Zn has been shown to significantly enhance the drought tolerance of plants, by improving their physiological and biochemical properties. This is achieved, in part, by strengthening the antioxidant defense system, which comprises enzymes such as catalase (CAT), peroxidase (POD), and superoxide dismutase (SOD), among others. These enzymes scavenge ROS and protect plant cells from oxidative stress.

In addition, Zn plays a critical role in maintaining cellular water balance by regulating the synthesis of osmoprotectants, such as proline and glycine betaine, which stabilize cellular structures and maintain turgor pressure in drought conditions (Siddiqui et al., 2022). Zn also helps reduce heavy metal toxicity by decreasing the uptake and accumulation of heavy metals such as cadmium (Cd) and lead (Pb), through modulation of membrane transporters and increasing the plant's antioxidant capacity. Zn maintains membrane stability by preserving lipid bilayer integrity and reducing lipid peroxidation (Hassan et al., 2022). Furthermore, Zn increases nutrient absorption by regulating the expression of genes involved in nutrient transport and assimilation and boosts the activity of antioxidant enzymes that detoxify heavy metals and protect cell components from damage (Syta et al., 2013). Zn is a vital component of various metalloenzymes involved in protein, nucleic acid, and chlorophyll synthesis, and influences the metabolism of carbohydrates, proteins, and auxins necessary for cell division and elongation (Suganya et al., 2020). To increase plant Zn use efficiency, strategies such as improving root architecture and uptake mechanisms can be employed through breeding programs aimed at creating Zn-efficient crop varieties and the application of Zn fertilizers and soil amendments that increase Zn bioavailability. Zn is not only vital for plant physiology but also plays a significant role in human nutrition, as it indirectly supports sustainable food production by ensuring an adequate intake of essential nutrients in the human diet (Tripathi et al., 2015). A deficiency of Zn in the soil can result in crops with insufficient Zn, leading to malnutrition and health issues in humans (Prahara et al., 2021). Therefore, boosting plant Zn efficiency is crucial for addressing global food security issues and improving human health (Verma et al., 2021). Enhancing plant Zn efficiency can improve crop resistance to environmental stresses, increase yields and nutritional quality, and contribute to the sustainability of food production systems (Hacisalihoglu, 2020).

Synthesis of ZnO NPs

NPs to be used as fertilizers can be synthesized by various methods like physical, chemical, and green synthesis by top-down and bottom-up approaches (Zulfiqar et al., 2019; Ghouri et al., 2020). Different physical, chemical, and biological (green) methods are used, depending upon the type of nanomaterial required and morphology of the materials being synthesized (Fig. 3).

Coating of ZnO NPs

For the desired activity of NPs, there is pressing demand for their

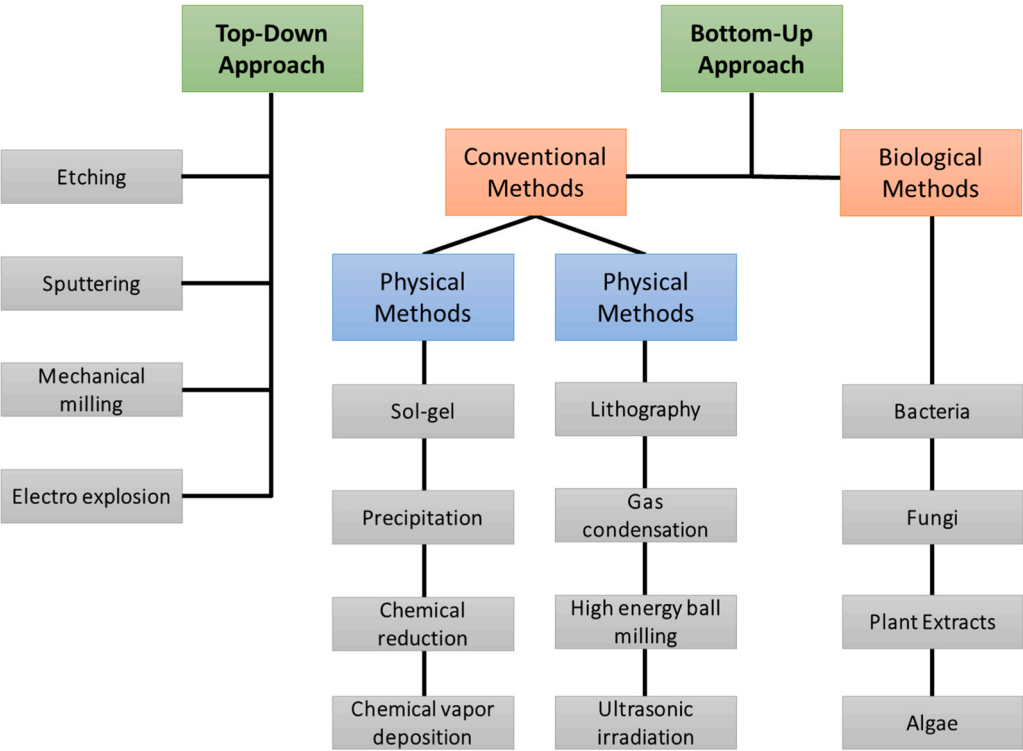


Fig. 3. Synthesis strategies used in top-down and bottom-up approaches.

coating. Coating NPs to improve their surface chemical and physical characteristics is critical for their reactivity (Da Silva et al., 2019). Coated NPs are also known as nanocomposites. It is a material made of several phases in which at least one, two, or even three dimensions are of nanometer scale range i.e., less than 100 nm. These structures have nanoscale repeat spaces between different phases that are forming the composite. This can include gels, colloids, copolymers, and porous media, but is more likely to contain the solid combination of bulk matrix and phases having nano dimensions, however, differing in properties (Rane et al., 2018). Constituents present in greater quantity are called matrix. Reinforcement process takes place by addition of constituents into matrix material, thus improving the properties of nanocomposites. The fillers of nanocomposite matrix incorporated are based on properties that need to be improved. The reinforcing fillers and matrix can be

inorganic or organic. This forms the basis for the classification of nanocomposites, hence classified as (i) metal matrix nanocomposites, (ii) polymer matrix nanocomposites, and (iii) ceramic matrix nanocomposites.

Nanocomposites are synthesized using multiphase solid materials that are usually a combination of two or even more components having a discontinuous or continuous phase. They are formed of a matrix in which various fillers are integrated. Owing to the importance of nanocomposite structures, various studies are aimed at improving their synthesis process. Thus, numerous mechanisms and improvements in their synthesis route are reported, most common of which as reported by Rane et al. (2018) are mentioned in Fig. 4. Nanocomposites are the materials of future due to their advantages over prior composite materials, some main privileges include a) high surface to volume ratio that

Coating Approaches for Synthesis of Nanocomposites		
Metal Nanocomposites	Ceramic Nanocomposites	Polymer Nanocomposites
• Spray pyrolysis • Rapid solidification • Chemical/Physical vapor deposition • Sol gel • Colloidal method	• Sol gel • Polymer precursor • Powder process	• Solution blending • Melt blending • In situ formation • Sol gel

Fig. 4. Coating approaches for synthesis of nanocomposites.

allows smaller filler size and the distance between fillers, b) enhanced optical properties such as light transmission, and c) upgraded mechanical properties like high ductility exclusive of strength loss and scratch resistance.

Engineered NPs such as ZnCuFe-O NPs, ZnO NPs, CaP NPs, FeO NPs, and Ag NPs have been shown in studies to promote plant growth in a variety of food crops when employed as micronutrient fertilizers. These crops include *Brassica species* (Mousavi Kouhi et al., 2015), *Solanum lycopersicum*, etc. (Shankramma et al., 2016). Phytochemical compounds extracted from seaweed (*Gracilaria gracilis*, *Cystoseira barbata*) can also enhance the germination of seeds, chlorophyll content, seedling growth, and yield in many crops (Khan et al., 2009). Therefore, coating of NPs with phytochemicals or other coating agents has been found beneficial for growth and development of the plants. Conjugation of NPs with coating material enhances the osmotic adaptation and crop water balance and leads to mitigation of stress-induced ROS damage with the aggregation of osmolytes. This results in enhanced morphological and physiological parameters in stress settings (Král'ová and Jampflek, 2023).

Effect of coated ZnO NPs on plants

Nanocomposite-based fertilizers (NCFs) have gained a lot of attention in agrochemical sector due to their multidimensional applications, biodegradability, and hydrophilic nature. NCFs are also referred to as smart fertilizers owing to their remarkable features such as efficient and controlled nutrient delivery without drastically affecting the surroundings. However, there is a need to conduct detailed experiments to correctly evaluate the pattern of their release and interaction with environmental components (Chakraborty et al., 2023). The combined effect of NPs and capping agents has proven to be a huge success depending on plant species and concentration of nanocomposites applied. Employing nanocomposites for preparing smart fertilizers results in a sustained, controlled, and coordinated supply of nutrients along with reduction in the chances of contamination (Kalia et al., 2020).

NPs can be surface coated with several agents to attain desirable traits within plants by soil or foliar application. Due to the enhanced efficacy and capacity to regulate in accordance with the circumstances

in which they are released, they are usually required in small quantities and are helpful in reducing environmental pollution (Kaur and Gupta, 2009). Coating agents serve as a stabilizer and enhance the biocompatibility of NPs within the living cells (Javed et al., 2020). Coating of osmoregulators like proline and betaine enhances plant growth by stabilizing sub-cellular structures like membranes and proteins, regulating ion channels, acting as radical scavenger, and increasing plant antioxidant defense system (Hanif et al., 2023; Hanif and Zia, 2023). Coating oleylamine on NPs improves photosystem II efficiency, hence acting as photosynthetic biostimulants for crop improvement (Tryfon et al., 2023). Urea coating improves Zn uptake in plants and maintains water balance, hence improving growth parameters (Dimkpa et al., 2020). Coating reduces the size of NPs, thus enhances reactivity of NPs. Coating with PEG and PVP elicits steviol glycosides formation, increasing biological activity, hence improves plant growth (Javed et al., 2017b). Similarly, other coating agents like proteins, amino acids, and polysaccharides improve plant growth by improving plant metabolic and photosynthetic efficiency, nutrient uptake, and regulating activities of enzymes for overcoming stress faced by plants (Kalimuthu et al., 2023; Jiménez-Rosado et al., 2021). Hence, the use of nanocomposites in the agricultural field is gaining a lot of attention due to its significant impact on morphological and nutritional traits along with improving stress tolerance within various plant species (Guha et al., 2020). The employment of different compounds as capping and reducing agents in the bioengineering of ZnO NPs among many other NPs is necessitated as shown in Table 1.

Uptake of ZnO NPs by plants

The uptake of Zn from soil to plant roots is predominantly in the form of divalent cation (Zn^{+2}). Plant tissues can be invaded by ZnO NPs through various entry points such as root or aerial tissues and other organs such as stomata and cuticles. The cell wall plays a vital role in regulating the uptake of NPs by filtering them according to size and shape. 5–20 nm is the size exclusion limit which prevents the entry of larger nanoparticles within the plant (Wang et al., 2016). Achari and Kowshik (2018) reviewed that the first defensive barrier in aerial plant parts is the cuticle which allows the passage of NPs smaller than 5 nm. On the other hand, NPs can enter through stomata having pore openings

Table 1
Effects of coated ZnO NPs on plants.

S. No.	Coated material	Concentration of NPs	Mode of application	Effects	Reference
1.	Phycomolecules	25–200 mg/L	Hydroponic	Increased biomass and chlorophyll content by 130.6 % and 141 % compared to control, respectively.	Venkatachalam et al. (2017)
2.	Oleylamine	15 mg/L	Foliar	Photosynthetic bio-stimulant for enhancing crop yield and electron transport rate by 20 %.	Tryfon et al. (2023)
3.	Urea	2.17 mg/kg	Soil	Enhanced plant performance and Zn accumulation under drought stress.	Dimkpa et al. (2020)
4.	Soy protein	0 wt%, 1.0 wt%, 2.0 wt%, 4.5 wt%	Soil	Nanofertilization in horticulture	Jiménez-Rosado et al. (2021)
5.	Alginate beads	100 mg/kg	Soil	Controlled release of Zn in agricultural soils	Martins et al. (2020)
6.	PEG and PVP	1 and 10 mg/L	MS Media	Increased biomass, protection against ROS damage, and improved physiological parameters at 1 mg/L.	Javed et al. (2017b)
7.	Cysteine	250 ppm	Seed priming	Enhanced germination of the plant (90 % and 76 %), radicle growth, seedling vigor, dry matter production, and antioxidant enzyme activity under normal and moisture stress conditions.	Kalimuthu et al. (2023)
8.	Chitosan	1 mg/L	MS media	Improved chlorophyll level by 51 % and doubled the amount of proline along with increase in carotenoids, alkaloids, and proteins.	Asgari-Targhi et al. (2021)
9.	Humic acid	—	Hydroponic	Increased germination (83 %), root length (60 %), shoot length, fresh weight, and dry weight.	Pirzada et al. (2022)
10.	Dextran	100,500 and 1000 mg/L	Hydroponic	Increased Zn concentrations in wheat without decreasing growth.	Elhaj Baddar and Unrine (2018)
11.	Maltodextrin (MDX)	1500 ppm	Foliar and drench	Increased plant height, stem diameter, and dry weight of plant organs (leaves, stem, and roots).	Pérez Velasco et al. (2020)
12.	Proline	50 and 100 mg/L	MS Media	Increased shoot length and root length under PEG-induced drought stressed plants.	Hanif et al. (2023)
13.	Betaine	50 and 100 mg/L	MS Media	Increase in morphological parameters under salt stress. At 100 mg/L, decreased SOD by 27 %, while POD by 33 %.	Hanif and Zia (2023)

larger than 20 nm and navigate towards phloem from where they are transported to stem, roots, and grains.

When added in soil, the NPs penetrate the root through plasmodesmata and symplastic routes and attach to carrier proteins or form new pores by endocytosis. However, in foliar application, NPs navigate through stomata or trichomes found in leaves (Fig. 5). Zn^{2+} ions commute through root epidermis, cortex, to endodermis, pericycle, xylem and then translocated through stem, leaves, to phloem, and seeds (Hacisalihoglu, 2020). Zn is predominantly absorbed by plants through roots by facilitated diffusion, mediated by transporter proteins and membrane potential. They can get past the plant defenses via apoplastic and symplastic routes to get to vascular system for upward transportation. However, ligand Zn complexes are also taken up by the roots of plants (Hamzah Saleem et al., 2022). Plants use two distinct strategies for Zn uptake depending upon ligands secreted by plant roots. The first strategy involves efflux of reductants as H^+ ions and organic acids that increases the solubility of Zn complexes (zinc hydroxides and phosphates), this results in the release of Zn ions for absorption by root epidermal cells. The second approach involves efflux of low molecular weight phytosiderophores that form stable complexes with Zn, fostering Zn influx into root epidermal cells (Umair Hassan et al., 2020). However, charged Zn ions are unable to cross root plasma membrane freely, therefore a specific set of transporter proteins are found involved. These include HMA family (heavy metal ATPases), ZIP family (zinc iron permease), MTP family (metal tolerance protein), VIT family (vacuolar iron transporter), CDF family (cation diffusion facilitators), and PCR family (plant cadmium resistance) proteins (Hacisalihoglu, 2020). After passing through casparian strips of roots, Zn ions enter cells of pericycle and xylem parenchyma by symplastic pathway. HMA transporters in xylem parenchyma play a major role in this transport. In xylem sap,

metal ions including v ions are transported as metal complexes with organic acids, asparagine, and nicotian amine histidine. After entering phloem, Zn is translocated to various plant organs. Zn mobility is higher in phloem as compared to xylem, because of greater solute concentrations like peptides and organic acids in phloem sap. Although xylem has lower solute concentration, it still functions fundamentally for nutrient transfer to different plant organs (Umair Hassan et al., 2020). ZnO NPs encounter different barriers within the plant cells which are affected by morphological traits of NPs, duration of exposure, and growth stage of host plant (Mousavi Kouhi et al., 2015).

ZnO NPs as nanofertilizers

NPs can bind and cargo different molecules like nutrients for their controlled release in plants (Dey et al., 2018; Meghana et al., 2021; Javed et al., 2023b). This slow release of molecules for longer time can minimize the risk of nutrient loss with environmental safety (Subramanian et al., 2015; Basit et al., 2022). Different metal oxide NPs have been reported to boost production of crops along with resistance to abiotic stress such as drought. Among many metal oxide NPs, TiO_2 , ZnO, and Fe_3O_4 NPs are frequently utilized in agriculture. These NPs have shown positive effects on crops grown under drought stress (Alabdallah et al., 2021). NPs as nanoscale agrochemicals, nanofertilizers, and nanoherbicides also contributed to the enhancement of agriculturally important crops (Singh et al., 2021). These nanoscale materials increase the solubility, controlled release, and targeted bio-distribution of active chemicals, resulting in fewer negative effects on non-target species resulting in lowering of environmental pollution (Agrahari and Dubey, 2020).

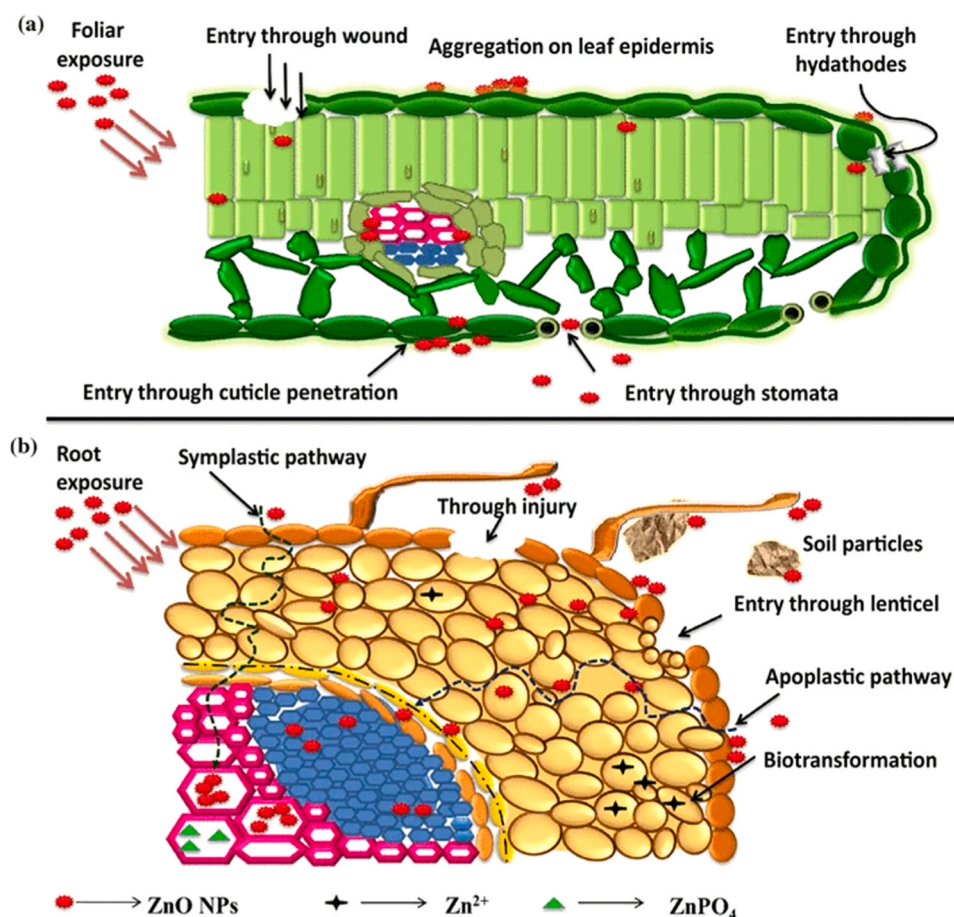


Fig. 5. An overview of uptake and translocation pathway of ZnO NPs in a plant system via a) foliar exposure, and b) root exposure (Singh et al., 2018).

In vitro applications of ZnO nanofertilizers

ZnO NPs used as fertilizers make 50–80 % Zn available to the plant. Different parameters affect the availability of Zn-based fertilizers. Various experiments have been conducted which confirmed that ZnO when applied in nano form is more potent as compared to bulk ZnO. A study conducted by Singh et al. (2013) proved this by investigating different biochemical parameters involved in seed germination and growth of cabbage, tomato, and cauliflower. They found out that ZnO NPs increased seedling growth, seed germination, and content of sugar, different pigments like chlorophyll, as well as activities of antioxidant enzymes like nitrate reductase. Adhikari et al. (2015) conveyed that plant growth parameters like root length and volume, plant biomass, and height of maize plant were all improved in plants that were applied with Zn as ZnO NPs as compared to ZnSO₄ in bulk.

In vitro application of ZnO NPs is performed by different modes. ZnO NPs can be incorporated by the process of seed priming, added to nutrient solution in case of hydroponic system, and/or added to culture media in case of tissue culturing. Each of these is found to show different effectiveness level. This is elaborated by the examples from literature in Table 2.

In vivo applications of ZnO nanofertilizers

ZnO NPs are capable of increasing plant yield and stress resistance by slowly releasing Zn ions into the plant (Sebesta et al., 2021). Comparatively, the use of bulk ZnO has been proven to negatively affect the plant biomass and biochemical processes except for enhancing root growth at higher concentrations (Nemček et al., 2020). In the study of Azarin et al. (2023), it was found that exposure to ZnO NPs led to a better impact on the photosynthetic efficiency in barley compared to bulk ZnO. Two modes of *in vivo* application of ZnO NPs have been studied extensively which includes supplementation by foliar spray and soil fertilization. The potential use of ZnO NPs as soil fertilizer seems to have beneficial effects on plant growth and productivity. Similarly, foliar spraying of ZnO NPs has also been proven as an efficient way to improve plant morphological parameters and physiological responses under different stress conditions (Lv et al., 2023). Both foliar and soil application of ZnO NPs can positively influence plant yield and stress tolerance, but careful selection of concentration and environmental aspects is necessary. However, the effect of ZnO nanofertilizers is influenced by multiple factors such as plant species, dosage, duration of treatment, and soil pH, etc (Zoufan et al., 2020). Hussain et al. (2018) reported ZnO NPs application to soil that reduced Cd concentration in the grains by 16–78 %, whereas foliar fertilization resulted in 30–77 % decline in Cd concentration within wheat roots, grains, and shoots as compared to the control. Moreover, ZnO NPs significantly improved the photosynthetic rate, Zn content, and grain yield of wheat. The foliar application of ZnO NPs has the potential for remediation of Zn-deficiency symptoms. For instance, the findings of study conducted by Bala et al. (2019) indicated that ZnO NPs improved microbial count in soil, enzyme activities, and yield attributing traits in rice cv. PR 121 grown in Zn-deficient soil.

The Okapi cultivar treated with different concentrations of ZnO NPs (0, 20, 40, and 80 mg L⁻¹) impacted the plant growth factors under salinity stress (Akhavan Hezaveh et al., 2020). It was reported that ZnO NPs at a concentration of 80 mg L⁻¹ promoted phenolic content and decreased Zn, P, and anthocyanin content. At 20 mg L⁻¹, seed oil content and seed weight were significantly increased (20.82 % and 79.28 %, respectively) at 50 mM NaCl stress. Mahdiah et al. (2018) studied the influence of nanoscale ZnO particles, Zn chelate, and ZnSO₄ on the growth of pinto bean cultivars by seed and foliar application. The authors reported that ZnO NPs at 15 % (w/v) enhanced shoot and internode length as well as yield of the plant as compared to the varying concentrations of other Zn materials being tested. In this study, foliar spraying proved to be more effective than seed application. Foliar

spraying resulted in improved plant growth parameters and high Zn content within the grains. The various effects of ZnO NPs in different plant species at different concentrations have been summarized in Table 3.

Challenging points of ZnO NPs as nanofertilizers

Despite the exciting opportunities and advantages which nanotechnology has to offer, it is still slowly evolving in agro-food sector because much more study is required to investigate toxicity and transfer of nano-based materials under environmentally relevant conditions (Ashraf et al., 2021). The need of the time is to explore the applications of nano-based products in agriculture at a bigger level to deal with unforeseen side effects.

Although ZnO nanoparticles (ZnO NPs) have great potential as nanofertilizers due to their ability to boost plant growth and productivity, several challenges and potential risks must be addressed to ensure their safe and effective application in agriculture (Usman et al., 2020). Regulating nanomaterials, including ZnO NPs, is a complex and evolving field, with current regulatory frameworks often lacking specific guidelines for their use in agriculture, leading to uncertainties and potential risks (Ghosh et al., 2022). It is crucial to develop comprehensive regulations for the production, application, and disposal of ZnO NPs that ensure safe handling of nanomaterials, establish permissible exposure limits, and mandate thorough testing of environmental and human health impacts (He et al., 2019).

It has been stated in various studies that exposure to ZnO NPs enhanced plant growth at low concentrations but negatively affected plant's yield at high concentrations, which might be caused by bioavailable Zn toxicity (Verma et al., 2021). Only a small quantity of Zn (0.05 mg/L) present in soil is needed for normal growth of plants and higher levels can have toxic consequences. The damage caused to a plant due to toxic levels of nanomaterials is referred to as nano-phytotoxicity. It is dependent on multiple factors such as NPs size, exposure duration, and soil pH. Another reason can be the release of Zn⁺² ions due to particle dissolution within the nutrient solution (Singh et al., 2018). At high concentrations, NPs can damage plant metabolism by employing various mechanisms which are a) oxidative damage induced by increased ROS production, b) genotoxicity due to interaction between organelles or DNA and NPs, c) interfering transcriptional processes and cell metabolism.

The long-term environmental impact of ZnO NPs is not fully understood. Once applied in agricultural fields, ZnO NPs can interact with soil components, water, and microorganisms, potentially leading to unintended consequences. Research has shown that ZnO NPs can undergo transformations, such as dissolution and aggregation, which can affect their bioavailability and toxicity (Sabir et al., 2014; Manchikanti and Bandopadhyay, 2010). Therefore, extensive research is needed to understand the long-term environmental footprint of ZnO NPs, including their persistence, mobility, and potential for bioaccumulation in the food chain (Murali et al., 2022).

Moreover, the long-term effects of ZnO NPs on soil health, plant systems, and non-target organisms remain largely unknown (Pérez-Hernández et al., 2020; Còa et al., 2020). Continuous exposure to ZnO NPs can lead to soil contamination, affecting soil microbial communities and ecological balance. Long-term studies are required to evaluate the cumulative impacts of ZnO NPs on soil fertility, plant health, and broader ecosystems. These studies should also assess the development of resistance in pests and pathogens exposed to ZnO NPs (Manzo et al., 2011; Kocbek et al., 2010).

One of the significant advantages of using ZnO NPs as nanofertilizers is their potential for controlled release of nutrients (López-Valdez et al., 2018). Although considerable progress has been made in the field of ZnO NPs, achieving precise control over the release rates remains a formidable challenge Raha and Ahmaruzzaman (2022). The inconsistent or uncontrolled release of ZnO NPs can lead to nutrient imbalances, which

Table 2*In vitro* effects of ZnO nanofertilizers on plants.

S. No.	Plant	Size of ZnO NPs	Method of synthesis	Concentration of ZnO NPs	Mode of application	Effects	Reference
1	Tomato (<i>Solanum lycopersicum</i> L.)	-	Chemical	30 mg/L	MS media	Increased callus formation even under salinity stress. Highest shoot formation per explant (12.5). Increased shoot regeneration to explant ratio, under salinity stress.	Aazami et al., 2021
2	Pomegranate (<i>Punica granatum</i> L.)	<100 nm	Chemical	0, 1, 2.5, 5, 7.5, and 10 mg/L	Woody plant medium (WPM)	Induced best proliferation rate at NPs concentration 2.5 mg/L. Maximum shoot length and biomass at 5 mg/L NPs concentration. Highest total chlorophyll content at 1 and 2.5 mg/L NPs concentration. Concentration increase up to 10 mg/L or more did not encourage plant growth. Proline buildup increased at 10 mg/L NPs concentration, while at 2.5 mg/L, proline accumulation radically reduced.	El-Mahdy and Elazab (2020)
3	Wheat (<i>Solanum lycopersicum</i> Mill. cv. royal)	15 nm	Chemical	0, 50, 100, 200, 400, 800, and 1600 mg/L	Seeds were positioned in petri dishes with filter paper and ZnO NPs suspension added	Seed germination tolerance index (GTI) and Root tolerance index (RTI) were positively influenced by 50 mg ZnO/L, and reduced at 100 mg/L. 50 mg ZnO/L clearly increased fresh weights of roots and shoots, while reduced with the increase in NP concentration ≥ 100 mg/L. 50 mg ZnO/L increased Chlorophyll a, b, and carotene content while reduced the photosynthetic pigments at 100 mg/L. MDA content increased at 200 mg ZnO/L. Proline content boosted in leaves at 200 mg ZnO/L. 100 mg ZnO/L enhanced antioxidant enzymes activity (POD, APX, SOD).	Amooaghaie et al. (2017)
4	Maize (<i>Zea mays</i> L.)	100 nm	Green	200 and 400 ppm	In petri plates (on germination sheets)	200 ppm ZnO NPs was found to be the optimum concentration as it increased the early seed germination %, root and shoot length, and fresh biomass. Greatest seed vigor index (SVI) was found at 200 ppm. No changes in young leaves emergence was found and 400 ppm reduced leaf number.	Sharma et al. (2022)
5	Onion (<i>Allium cepa</i> L. 'Sochaczewska')	$\cong 20$ nm to $\cong 60$ nm	Chemical	50, 100, 200, 400, 800, 1600, and 3200 mg/L	In MS media (with some modifications)	Highest germination rate was documented at 800 mg/L ZnO NPs, i. e., 56 %. 3200 mg/L showed lowest germination rate (11 %). 50–1600 mg/L NPs was suitable for seed germination.	Tymoszuk and Wojnarowicz (2020)
6	<i>Stevia rebaudiana</i> Bertoni	34 nm	Chemical	0.1, 1.0, 10, 100, 1000 mg/L	In MS media	Increase in growth parameters, steviol glycosides (rebaudioside A and stevioside) production, and antioxidant activities in the tissue culture grown shoots	Javed et al. (2017a)
7	Tobacco (<i>Nicotiana tabacum</i>)	25 nm	Chemical	0.015, 0.03, 0.06, 0.12, and 0.24 mM	In MS media	Resulted in increased callus zinc content, and callus DW and FW at 0.12 and 0.24 mM. 0.03 mM enhanced protein callus content that was linked positively with ROS and MDA contents, while SOD enzyme activity notably decreased at all concentrations.	Mazaheri-Tirani and Dayani 2020
8	Eggplant (<i>Solanum melongena</i>)	18 nm	Chemical	100, 250, 500, and 1000 mg/L	In culture vessels covered with filter paper	Total chlorophyll content, and shoot and root length sequentially declined with ZnO NPs concentration. A subsequent increase in MDA content was observed.	Baskar et al. (2018)

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Table 2 (continued)

S. No.	Plant	Size of ZnO NPs	Method of synthesis	Concentration of ZnO NPs	Mode of application	Effects	Reference
9	Broccoli (<i>Brassica oleracea</i> var <i>italica</i>)	24 nm	Green	0, 50, 100, 200, 400, 800, and 1000 µg/L	In petri dishes with the application of NPs suspension	Improved anthocyanin was observed (up to 500 mg/L). DPPH activity was more evident at lower concentrations (100 and 250 mg/L). TPC and TFC increased up to (500 mg/L). DNA damage was detected in all treatments. 37.5 % enhancement in seed germination. Notable increase in root length and weight of seedlings. Leaf number, plant height, and leaf area increased (at optimum concentration 800 µg/L). Proline, phenolic, sugar, and chlorophyll content was increased remarkably (at optimum concentration 200 µg/L).	Awan et al. (2021)
10	Chinese cabbage (<i>Brassica pekinensis</i> L.)	30 nm, 50 nm, 90 nm, and 150 nm	Chemical	0, 1, 5, 10, 20, 40, and 80 mg/L	In petri plates with filter paper and NPs suspension added	90–100 % germination rate. Shoot and root elongation found to be inhibited. Zinc uptake was higher by roots than shoots.	Xiang et al., 2015
11	Garden cress (<i>Lepidium sativum</i>)	48.9 nm, 37.4 nm, and 43.6 nm	Green	250, 500, 1000, 2000, and 4000 ppm	Seeds placed on petri dish and ZnO NPs solution added	Increased plumule and radicle lengths. Increased fresh and dry weights. Increased total chlorophyll and carotenoid content.	Sedefoglu et al. (2022)
12	Lettuce (<i>Lactuca sativa</i>)	50 nm	Chemical	0, 25 and 50 µg/mL	In petri dishes with soaked filter paper	Germination enhanced, biomass improved, Zn level boosted inside seeds. Higher capability to synthesize proteins. Stimulated catalase enzyme activity. Alleviated level of total carbohydrates in treated seeds.	Rawashdeh et al. (2020)
13	Green gram, Bengal gram, and Turkish gram	78–84 nm	Green	8.6 mg/L	In MS media	100 % germination rate. Increased protein and chlorophyll content. Turkish gram and Bengal gram roots took up ZnO NPs, while there was low translocation to shoots, and opposite scenario was observed in green gram.	Ukidave et al., 2021
14	Rice (<i>Oryza sativa</i> L.) Kargi and CSR 30 variety	-	Chemical	50 mg/L	In Borosil culture test tubes containing Hougland's solution with ZnO NPs	Mitigated salinity stress. Upregulated antioxidative activity (SOD, CAT, GR, and APX). Reduced proline, H ₂ O ₂ , and MDA level under stress. Accumulation of photosynthetic pigment and K ⁺ level. Reforming trichome and stomata morphology.	Singh et al. (2022)
15	Pea (<i>Pisum sativum</i> L.)	<50 nm and <100 nm	Chemical	100, 250, and 1000 mg/L	In petri dishes with NPs suspension	Improved completion of seed germination (at 100 mg/L). Slightly decreased elongation of epicotyl and roots. Increased polar metabolites in growing tissues. High sucrose particularly in epicotyl (at 100 mg/L) and roots (at 250 mg/L). Homoserine was dominant in all treatments. Highest level of mallic acid in epicotyl and roots (at 250 mg/L concentration).	Staśnawska et al. (2023)
16	Potato (<i>Solanum tuberosum</i> L.) Nicola, Hermes, and Spunta cultivars	30 nm	Chemical	50 mg/L	In MS media (meristem tip explants were planted)	Increased callus fresh weight even under salt stress. Calcium accumulation in potato tuber. NPs enabled tubers to retain Fe even under salinity stress.	Alghamdi et al. (2022)

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Table 2 (continued)

S. No.	Plant	Size of ZnO NPs	Method of synthesis	Concentration of ZnO NPs	Mode of application	Effects	Reference
17	Barley (<i>Hordeum sativum</i> L.)	15–30 nm	Chemical	300 and 2000 mg/L	Hydroponic system	Elevated MDA, CAT, and proline content in shoots. SOD activity increased by x2 at 300 mg/L and x3.3 at 2000 mg/L in roots. GST, APOX, and glutathione reductase activity boosted in roots. Zn content in aerial plant parts was x2 more at 300 mg/L concentration. ~3 % increase in shoot and root length was observed in concentration less than 10 mg/L. Higher concentration caused reduction in length. Zinc internally absorbed by seedlings was 6 times higher than that adsorbed on seed surface. ZnO NPs reduced the toxic impact of CuO NPs to the seedlings (binary NPs were used).	Voloshina et al. (2022)
18	Radish (<i>Raphanus sativus</i>)	<40 nm	Chemical	0, 0.1, 1, 10, 100, and 1000 mg/L	In petri plates with NPs suspension added	Longest roots observed at 50 mg/L (66 % increase). 40 % increase in length of shoots in both treatments. GPOX activity decreased in roots, while increased in leaves and stems. Activity of NADH oxidase increased in leaves, roots, and stems. Shoot number, FW, and DW boosted at 6 and 18 mg/L. Soluble proteins and chlorophyll a, b content enhanced at 6 and 18 mg/L. 2 mg/L had no enhancing effect. Carotenoids and anthocyanins were stimulated at 18 mg/L. 6 and 8 mg/L were effective in increasing antioxidant activity. Increased levels of POD, CAT, and proline at 15 mg/L concentration. Salinity stress alleviated. 15 mg/L increased shoot number, increased K ⁺ uptake, and decreased Na ⁺ /K ⁺ ratio even under salt stress (at 15 mg/L concentration). Increased leaf thickness (lamina, medvian, upper epidermis, spongy and palisade tissues thickness) even under salt stress.	Singh and Kumar 2019
19	Fenugreek (<i>Trigonella foenum-graecum</i> L.)	<50 nm	Chemical	5 and 50 mg/L	Seeds grown on wet filter paper and then transferred to Hoagland's solution having NPs	Shoot number, FW, and DW boosted at 6 and 18 mg/L. Soluble proteins and chlorophyll a, b content enhanced at 6 and 18 mg/L. 2 mg/L had no enhancing effect. Carotenoids and anthocyanins were stimulated at 18 mg/L. 6 and 8 mg/L were effective in increasing antioxidant activity. Increased levels of POD, CAT, and proline at 15 mg/L concentration. Salinity stress alleviated. 15 mg/L increased shoot number, increased K ⁺ uptake, and decreased Na ⁺ /K ⁺ ratio even under salt stress (at 15 mg/L concentration). Increased leaf thickness (lamina, medvian, upper epidermis, spongy and palisade tissues thickness) even under salt stress.	Chemingui et al. (2019)
20	Olive (<i>Olea europaea</i>) Moraiole cultivar	10–20 nm	Green	0, 2, 6, and 18 mg/L	Olive growth medium with agar, sucrose, and zeatin	Increased levels of POD, CAT, and proline at 15 mg/L concentration. Salinity stress alleviated. 15 mg/L increased shoot number, increased K ⁺ uptake, and decreased Na ⁺ /K ⁺ ratio even under salt stress (at 15 mg/L concentration). Increased leaf thickness (lamina, medvian, upper epidermis, spongy and palisade tissues thickness) even under salt stress.	Regni et al. (2022)
21	Strawberry (<i>Fragaria x ananasa</i> Duch.)	<30 nm	Chemical	0, 15, and 30 mg/L	In MS media	Increased levels of POD, CAT, and proline at 15 mg/L concentration. Salinity stress alleviated. 15 mg/L increased shoot number, increased K ⁺ uptake, and decreased Na ⁺ /K ⁺ ratio even under salt stress (at 15 mg/L concentration). Increased leaf thickness (lamina, medvian, upper epidermis, spongy and palisade tissues thickness) even under salt stress.	Zeid et al. (2023)

can adversely affect plant growth and soil health ([Rico et al., 2011](#); [Raliya et al., 2017](#)). Therefore, it is essential to direct research efforts towards developing advanced formulations and coatings that allow for targeted and sustained release of Zn, thereby ensuring optimal nutrient availability while minimizing the environmental impact ([Vejan et al., 2021](#); [Ahmed and Khurshid, 2011](#)).

Conclusions and future recommendations

Use of ZnO NPs and their derivatives as nanofertilizers has become a promising technique to optimize plant growth and development in agricultural practices. Zn, an important element for plants, is involved in a variety of physiological activities such as photosynthesis, nucleic acid metabolism, and hormonal control. ZnO NPs have unique properties such as increased surface area and reactivity, making them useful in improving nutrient uptake by plants. The small size allows for absorption through plant cell membranes, resulting in effective Zn delivery at the cellular level. Moreover, controlled release of ZnO NPs increases its availability in the soil, solving the problems associated with nutrient leaching and providing plants with a constant nutrient supply. Several investigations have found that ZnO NPs improve seed germination, root

development, chlorophyll content, and photosynthetic efficiency, resulting in healthier and stronger plants. Additionally, the use of ZnO NPs derivatives, viz., coated ZnO nanofertilizers to supplement different nutrient sources has also been reported. Combining nanotechnology with traditional Zn and other sources enables personalized strategies to meet the specific needs of various crops and soil types. As research in this sector continues to evolve, knowledge of the environmental impact and long-term applications will be critical for realizing the full potential of ZnO NPs in modern agriculture. Further research is indispensable in several critical areas to fully harness the potential of ZnO NPs as nanofertilizers. These areas include understanding the specific pathways through which ZnO NPs enhance plant growth and stress tolerance, determining the optimal concentrations and application methods for different crops and soil types, investigating how ZnO NPs interact with various soil components and plant systems, and exploring alternative nanomaterials that could complement or enhance the benefits of ZnO NPs, thereby providing a broader range of solutions to agricultural challenges.

Table 3
In vivo effects of ZnO nanofertilizers on plants.

S. No.	Plant	Size of ZnO NPs	Concentration of ZnO NPs	Mode of application	Effects	References
1.	Pepper (<i>Capsicum annum</i> L.)	12–30 nm	0, 1000, and 2000 ppm	Foliar	Enhanced the salt tolerance by stimulating the antioxidant defense system. Significant increase in proline content and reduction in electrolyte leakage.	Rasouli et al. (2022)
2.	Maize (<i>Zea mays</i>)	33 nm	40, 80, 120, 160 mg/kg and 10, 20, 30, 40 ppm	Soil and Foliar	Soil application caused 61.1 % increase in plant growth and 51.8 % rise in photosynthetic pigments. Foliar application resulted in enhanced plant growth by 59.28 %, and antioxidant activity by 52.91 %.	Azam et al. (2022)
3.	Crimson seedless grape (<i>Vitis vinifera</i> L.)	97.31 nm	0, 25, 50, 100, and 250 ppm	Foliar	ZnO NPs at 25 ppm led to enhanced red coloration and the highest enzyme activity (PAL and SOD). Fruit firmness, anthocyanin, and total soluble solids content increased, whereas the acidity in berries declined.	Abou El-Nasr et al. (2021)
4.	Soybean (<i>Glycine max</i> cv. Kowsar)	38 nm, 59 nm, and >500 nm	120 mg and 400 mg/kg	Soil	Enhanced the levels of MDA, SOD, CAT, and POX. At 400 mg Zn/kg, negative effects on plant growth parameters were observed.	Yusefi-Tanha et al. (2020)
5.	Tomato (<i>Lycopersicon esculentum</i> Mill.)	-	10, 50, and 100 mg L ⁻¹	Foliar	Regulated plant's response to salinity stress and boosted shoot length, root length, biomass, leaf area, and chlorophyll content.	Faizan et al. (2021)
6.	Bean seeds (<i>Phaseolus vulgaris</i> L.)	10–30 nm	0, 250, 500, 1000, and 2000 mg/L	Soil and Foliar	Foliar spray caused a significant impact on plant metabolism and yield, while soil irrigation elicited chlorophyll related proteins.	Salehi et al. (2021)
7.	Coffee (<i>Coffea arabica</i> L.)	68.14 nm	10 mg L ⁻¹	Foliar	Improved the fresh and dry weight of roots and leaves, and positively affected photosynthetic rate of the plant.	Rossi et al. (2019)
8.	Potato (<i>Solanum tuberosum</i> L.)	50 nm	0, 50, 100 ppm	Soil	Successfully mitigated drought stress at 100 ppm. Enhanced photosynthesis and productivity of the plant.	Al-Selwey et al. (2023)
9.	Cherry Tomato (<i>Solanum lycopersicum</i> L. var. cerasiforme)	<50 nm	03, 20, and 225 mg/kg	Soil	Plant yield enhanced up to 23 % in calcareous soil and 16 % in the acidic soil as compared to the control. However, exposure to higher concentration (225 mg/kg) negatively affected tomato yield.	Almendros et al. (2022)
10.	Pigweed (<i>Amaranthus</i>)	23.69 nm	10, 50, and 100 mg L ⁻¹	Foliar	Foliar application at 10 mg/L resulted in the highest plant height and better nutrient recovery. The protein and glucosinolates content was relatively higher in all treated plants. However, CuO NPs led to a greater seed yield and diameter in contrast to ZnO NPs.	Reshma and Meenal. (2022)
11.	Black mustard (<i>Brassica nigra</i>).	41 nm	200, 400, and 600 mg/kg	Soil		Zafar et al. (2020)
12.	Cluster-bean (<i>Cyamopsis tetragonoloba</i> L.)	3.8 nm	10 ppm	Foliar	Significantly improved the plant biomass and chlorophyll content along with its root and shoot length.	Raliya and Tarafdar. (2013)
13.	Maize (<i>Zea mays</i>)	30–70 nm	8 kg ha ⁻¹ and 2 % solution	Soil and Foliar	ZnO NPs successfully improved growth parameters and Zn content in edible portion of plant. This study demonstrated an insignificant difference in both application methods (soil and foliar).	Umar et al. (2021)
14.	Basil (<i>Ocimum basilicum</i> L.)	20–60 nm	5, 10, 15, and 20 mg L ⁻¹	Foliar	At 20 mg L ⁻¹ concentration, plants exhibited highest carotenoid and vitamin C content along with enhanced morphological parameters.	Fortis-Hernández et al. (2023)
15.	Lettuce (<i>Lactuca sativa</i> L.)	90 ± 10 nm	0, 1, 10, 100 mg ZnO/kg	Soil	Optimal dose of 10 mg/kg resulted in enhanced photosynthetic rate and biomass. Higher Zn content was displayed by the plants treated with ZnO NPs as compared to its counterpart (bulk ZnO) at similar concentrations.	Xu et al. (2018)
16.	Dragonhead (<i>Dracocephalum moldavica</i> L.)	10–30 nm	160 and 400 mg L ⁻¹	Foliar	ZnO NPs at the optimal concentration of 160 mg/L led to higher levels of secondary metabolites and essential oil yield.	Nekoukhou et al. (2022)
17.	Sorghum (<i>Sorghum bicolor</i> var. 251)	18 nm	1, 3, and 5 mg/kg	Soil	Higher nitrogen and phosphorus uptake by roots under drought stress was observed, regardless of the dose. The delay in grain head formation due to drought was reduced.	Dimkpa et al. (2019)
18.	Habanero pepper (<i>Capsicum hinese</i> Jacq.)	12–20 nm	1000 and 2000 mg/L	Foliar	1000 mg/L concentration of ZnO-NPs positively influenced fruit yield and nutraceutical properties of the plant. ZnO NPs at 2000 mg/L led to higher antioxidant	García-López et al. (2019)

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Table 3 (continued)

S. No.	Plant	Size of ZnO NPs	Concentration of ZnO NPs	Mode of application	Effects	References
19.	Green Pea (<i>Pisum sativum</i> L.)		250 and 1000 mg/kg	Soil	capacity and capsaicin content despite negatively affecting the plant's growth. Higher Zn accumulation was shown by coated ZnO NPs at 250 mg kg ⁻¹ . Zn exposure had non-significant impact on the protein and carbohydrate profile of plants.	Mukherjee et al. (2016)
20.	Rice (<i>Oryza sativa</i> L.)		0, 0.5, 1.0, 5.0 g L ⁻¹	Foliar	Promoted the microbial count and Zn content according to the concentration applied.	Bala et al. (2019)
21.	Cilantro (<i>Coriandrum sativum</i>)	24 ± 3 nm	100, 200, and 400 mg/kg	Soil	A rise of 50 % chlorophyll content as compared to the control was observed. At 400 mg/kg, ZnO NPs caused a decline in lipid peroxidation by 70 %.	Pullagurala et al. (2018)
22.	Eggplant (<i>Solanum melongena</i> L.)	-	0, 50, and 100 ppm	Foliar	Fruit yield was increased by 12.2 % and 22.6 % by foliar spraying of 50 and 100 ppm ZnO NPs, respectively. Photosynthetic rate and water productivity of water-stressed eggplants was significantly improved.	Semida et al. (2021)
23.	Barley (<i>Hordeum vulgare</i> L.)	-	100 mg/kg and 100 mg L ⁻¹	Soil and Foliar	Foliar supplementation of ZnO NPs was more effective in alleviating the oxidative stress caused by salt stress as compared to soil application.	Ali et al. (2022)
24.	Lettuce (<i>Lactuca sativa</i> L.) and Carrot (<i>Daucus carota</i> subsp. <i>sativus</i>)	< 50 nm	1, 5, 20, 100, and 1000 mg/kg	Soil	A significant increase in biomass of both plants was observed as compared to the control. However, 1000 mg kg ⁻¹ ZnO NPs negatively affected the growth of lettuce plants.	Song and Kim (2020)
25.	Rice (<i>Oryza sativa</i> L.)	50–500 nm	60 and 120 mg L ⁻¹	Foliar	Led to better yield performance among different rice varieties.	Somarathne et al. (2021)

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CRediT authorship contribution statement

Muhammad Zia: Writing – review & editing, Visualization, Validation, Supervision, Project administration, Methodology, Formal analysis, Conceptualization. **Snovia Farooq:** Writing – original draft, Investigation, Data curation. **Misbah Zeb Kiani:** Writing – original draft, Investigation, Data curation. **Mumtaz Cheema:** Writing – review & editing, Visualization, Supervision, Formal analysis. **Rabia Javed:** Writing – review & editing, Writing – original draft, Visualization, Formal analysis, Data curation. **Saad Hanif:** Writing – original draft, Investigation, Data curation, Conceptualization.

Declaration of Competing Interest

All authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Consent for publication

All authors agree to proceed with the manuscript for publication.

Authors' contributions

All authors contributed equally to the design and writing of the manuscript.

Data availability

No data was used for the research described in the article.

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